

Artificial Intelligence (CSC266)

Chapter-wise Study & Priority Guide

BSc. CSIT – 4th Semester – Tribhuvan University

Based on TU Exam Papers: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082, Model (2077 paper not available at time of analysis)

■ ×1 Low – appeared once	■ ×2 Medium – appeared twice
■ ×3–4 High priority	■ ×5+ MUST PREPARE

★ = extremely high priority — appeared in 5+ of 8 papers, prepare thoroughly with worked examples.

Unit 1 – Introduction (3 Hrs)

Defining AI / Turing Test (acting & thinking humanly and rationally) ■ ×8 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082, Model

Q: Define AI from a given perspective (thought process, behavioral process, rationality); explain the Turing Test and what properties an agent needs to pass it.

Foundations of AI (philosophy, psychology, economics, linguistics) ■ ×2

Years: 2078, 2081

Q: Discuss how disciplines like philosophy, sociology and economics influence the study of AI, or describe the foundations of AI in general.

■ **What to read:** Short unit but the AI/Turing Test definition question is the single most reliable opener across every paper — memorize 2–3 clean framings (acting/thinking humanly vs rationally) so you can adapt to whichever angle is asked. Don't skip this thinking it's "too basic to be asked" — it's asked every year.

Unit 2 – Intelligent Agents (4 Hrs)

PEAS Framework Design ■ ×5 ★

Years: 2078, 2080, 2080(New), 2082, Model

Q: Design a PEAS (Performance, Environment, Actuators, Sensors) description for a given intelligent agent (e.g. medicine delivery drone, internet shopping assistant).

Agent Types (simple reflex, model-based, goal-based, utility-based, rational agent) ■ ×4

Years: 2079, 2080, 2080(New), Model

Q: Differentiate between two agent types (e.g. utility-based vs model-based, simple reflex vs model-based) with an example.

Environment Types (deterministic, stochastic, static, dynamic, observable, single/multi-agent) ■ ×3

Years: 2076, 2081, 2082

Q: Justify which environment type resembles a given scenario, or discuss the types of environment an agent can work in, with examples.

■ **What to read:** PEAS is the go-to numerical-style design question — practice writing PEAS for 4–5 different agent types (drone, tutor, shopping assistant, diagnosis system) so you can adapt fast. Agent types and environment types are usually paired as a quick differentiate/justify question — keep a one-line definition per type ready.

Unit 3 – Problem Solving by Searching (9 Hrs)

Informed Search Numerical (Greedy Best-First, A*, Hill Climbing) ■ ×6 ★

Years: 2079, 2080, 2080(New), 2081, 2082, Model

Q: Given a state space with heuristic values, trace how Greedy Best-First Search, A* Search, or Hill Climbing expands nodes to reach the goal.

Adversarial Search Numerical (Minimax, Alpha-Beta Pruning) ■ ×6 ★

Years: 2078, 2080, 2080(New), 2081, 2082, Model

Q: Apply the Minimax algorithm or Alpha-Beta Pruning to a given game tree; show pruned branches and final alpha/beta values at each node.

Uninformed Search (DFS, BFS, Depth Limited, Iterative Deepening, Uniform Cost, Bidirectional) ■ ×6 ★

Years: 2076, 2078, 2079, 2080, 2081, 2082

Q: Illustrate how a given uninformed search strategy (Depth Limited Search, Iterative Deepening, Uniform Cost Search) finds a goal in a state space, including its benefits and limitations.

Constraint Satisfaction Problem (CSP) ■ ×2

Years: 2079, 2080(New)

Q: Define CSP and illustrate a problem (e.g. graph coloring) as a constraint satisfaction problem, or compare CSP with a real-world problem.

Search Completeness & Admissibility Concepts ■ ×2

Years: 2076, 2079

Q: Define admissible heuristic with an example; show why a given search strategy is or isn't complete for a constructed state space.

■ **What to read:** The three starred topics here are the backbone of the whole exam — minimax/alpha-beta and informed search node-tracing are close to guaranteed as long-answer questions. Practice tracing by hand on paper, not just reading the algorithm — graders check the trace, not the final answer alone.

Unit 4 – Knowledge Representation (14 Hrs)

FOPL Conversion & Resolution Inference ■ ×8 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2081, 2082, Model

Q: Convert a set of natural-language statements into First Order Predicate Logic (FOPL), then use the resolution algorithm to prove or infer a given conclusion.

Semantic Networks ■ ×4 ★

Years: 2078, 2080(New), 2082, Model

Q: Construct a semantic network for a given set of facts, or explain how knowledge is inferred from a semantic net.

Bayesian / Belief Networks & Probability Numerical ■ ×4 ★

Years: 2076, 2078, 2082, Model

Q: Given prior/conditional probabilities, construct a belief network or compute posterior probability using Bayes' Rule (e.g. probability of disease given a test result).

Fuzzy Logic ■ ×3

Years: 2076, 2079, 2081

Q: Define fuzzy logic and construct a fuzzy set or fuzzy rule-based expert system using given assumptions.

Frames ■ ×2

Years: 2076, 2080

Q: Define a frame and represent given knowledge (e.g. employee record) using a frame structure.

Scripts ■ ×1

Years: 2081

Q: Explain how knowledge is represented using scripts and create a small knowledge base using a script.

■ **What to read:** Heaviest unit by far (14 hrs) and it earns that weight — FOPL + resolution appears in *every single paper*, no exceptions. This is your highest-ROI topic: master the sentence-to-FOPL translation rules and the resolution proof steps until they're automatic. Semantic nets and Bayesian networks are the next tier — practice constructing both from a written scenario, since that's always how they're asked.

Unit 5 – Machine Learning (9 Hrs)

Genetic Algorithm (selection, crossover, mutation operators) ■ ×6 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2082

Q: Define the genetic algorithm operators (selection, crossover, mutation); given chromosomes, show the result of one-point or two-point crossover.

Artificial Neural Networks (structure, backpropagation, perceptron, Hebbian learning) ■ ×5 ★

Years: 2076, 2078, 2080, 2081, Model

Q: Define the mathematical model of an ANN; discuss how backpropagation, perceptron learning, or Hebbian learning is used to train a neural network.

Reinforcement Learning ■ ×3

Years: 2079, 2081, 2082

Q: Define reinforcement learning; differentiate active vs passive reinforcement learning, or model-free RL.

Naive Bayes Model / Supervised vs Unsupervised Learning ■ ×2

Years: 2080(New), Model

Q: Differentiate supervised learning from unsupervised learning; discuss how the Naive Bayes model works with an example.

Hebb Net / Perceptron Numerical Design ■ ×2

Years: 2081, 2082

Q: Design a Hebb net or configure an ANN neuron/perceptron to simulate a given logical function (e.g. OR gate).

■ **What to read:** GA operators and ANN training methods are the two pillars — both show up as either theory or a small worked example (crossover on given chromosomes, a single backprop iteration). Know the GA operator definitions cold and be able to walk through one backpropagation/perceptron update step by hand.

Unit 6 – Applications of AI (6 Hrs)

Natural Language Processing (steps, syntactic/semantic/pragmatic analysis, NLG vs NLU) ■ ×6 ★

Years: 2076, 2078, 2079, 2080, 2080(New), 2082

Q: Discuss the steps of Natural Language Processing; explain how syntactic, semantic or pragmatic analysis is performed, with an example.

Expert Systems (development phases, components, inference engine) ■ ×5 ★

Years: 2078, 2079, 2080, 2082, Model

Q: Define expert system with example; explain the phases of expert system development and the role of the inference engine.

Machine Vision & Robotics ■ ×3

Years: 2076, 2080(New), 2081

Q: Explain how machine vision is used in robotics, or describe the components of a machine vision system.

■ **What to read:** NLP's three analysis stages (syntactic, semantic, pragmatic) are asked almost interchangeably — know what each stage does and have one clean example ready. Expert systems is a safe, mostly-definitional topic — phases of development + inference engine role covers most of what's asked.

Top Priority Summary (Sorted by Frequency)

Topic	Unit	Freq	Years
FOPL Conversion & Resolution Inference	Knowledge Representation	×8	2076–82,Model
Defining AI / Turing Test	Introduction	×8	2076–82,Model
Informed Search Numerical (A*/Greedy/Hill Climbing)	Problem Solving by Searching	×6	2079–82,Model
Adversarial Search Numerical (Minimax/Alpha-Beta)	Problem Solving by Searching	×6	2078,80,80(N),81,82,Model
Uninformed Search (DFS/BFS/DLS/IDS/UCS)	Problem Solving by Searching	×6	2076,78,79,80,81,82
Genetic Algorithm (operators)	Machine Learning	×6	2076,78,79,80,80(N),82
Natural Language Processing	Applications of AI	×6	2076,78,79,80,80(N),82
PEAS Framework Design	Intelligent Agents	×5	2078,80,80(N),82,Model
Artificial Neural Networks (backprop/perceptron/Hebbian)	Machine Learning	×5	2076,78,80,81,Model
Expert Systems	Applications of AI	×5	2078,79,80,82,Model
Agent Types (reflex/model/goal/utility)	Intelligent Agents	×4	2079,80,80(N),Model
Semantic Networks	Knowledge Representation	×4	2078,80(N),82,Model
Bayesian / Belief Networks & Probability	Knowledge Representation	×4	2076,78,82,Model
Environment Types	Intelligent Agents	×3	2076,81,82
Fuzzy Logic	Knowledge Representation	×3	2076,79,81
Reinforcement Learning	Machine Learning	×3	2079,81,82
Machine Vision & Robotics	Applications of AI	×3	2076,80(N),81

Final Exam Very High Probability Topics (Never Miss These)

MUST PREPARE (×5+, near-certain): FOPL & resolution, Defining AI/Turing Test, Informed search numerical, Adversarial search (minimax/alpha-beta), Uninformed search, Genetic algorithm operators, NLP, PEAS framework, ANN training, Expert systems.

High priority (×3–4): Agent types, Semantic networks, Bayesian/belief networks, Environment types, Fuzzy logic, Reinforcement learning, Machine vision & robotics.

Numerical / worked-example heavy (practice by hand): Alpha-beta pruning traces, A*/greedy/hill-climbing node expansion, minimax game trees, FOPL-to-resolution proofs, Bayes' rule probability, GA crossover on given chromosomes.

Due for a repeat (asked before, quiet recently — good bet): Frames, Scripts, Naive Bayes model, CSP/graph coloring, Hebb net design.

